

BE Take Home Test - Candidate Information

Assignment [🔗](#)

Task summary [🔗](#)

The task is to develop an API that demonstrates API throttling, you will implement two `GET` endpoints: `/foo` and `/bar`. Each of these endpoints will include rate limiting to regulate the number of requests permitted within a defined time frame. We recommend you spend around 4 hours on this task.

Requirements [🔗](#)

- Your submission should compile and have instructions on how to run it defined in the `readme.md`
- No restrictions on framework or language use whatever you are comfortable with.
- Implement your own rate limiting logic.
- All request will have authentication in the form of an Authorization header containing a clientID to authorise a specific client. this has the following format `Authorization : bearer <client-id>`.
- `/foo` and `/bar` should leverage different rate limiting algorithms.
- Both endpoints should return `200 - { succes:true }` when rate limit is not reached.
- Both endpoints should return `429 - { error: 'rate limit exceeded'}` when rate limit is reached.
- Each client (defined by the `Authorization` header) should have a configurable rate limit.
- We want at least 2 supported clients with different rate limits.
- We want you to implement 2 storage strategies for the rate limit counters. 1 in memory and 1 in persistent storage.
- Be prepared to demonstrate the working functionality for at least 2 clients calling both endpoints with both storage strategies.
- Your solution should contain at least 1 test.
- Your solution is deployed on a Cloud provider and accessible for us to test **(stretch)**

Submission [🔗](#)

You can create a public or private github repo with your submission and

- In case of a public repo, send a link to your recruiter
- In case of a private repo, invite the ShowpadInterviewer(TBD) user.

Next steps [🔗](#)

We will review your submission, if it passes our code review, we will invite you to a follow-up interview were you will be asked to walk us through your implementation, elaborate on design and algorithm choices. We will also ask you to add more features to the project so be prepared to walk us through your project in an editor of your choice

Extra resources [↗](#)

Example of a **non throttled** API request with the `curl` command on the `/foo` endpoint

```
1 // Simulating request to /foo
2 > curl -X GET "http://api.example.com/foo" \
3 >   -H "Authorization: Bearer client-1"
4 < HTTP/1.1 200 OK
5 < Content-Type: application/json
6 < {
7 <   "success": true
8 < }
```

Example of a **throttled** API request with the `curl` command on the `/foo` endpoint

```
1 // Simulating request to /foo
2 > curl -X GET "http://api.example.com/foo" \
3 >   -H "Authorization: Bearer client-1"
4 < HTTP/1.1 429 TOO MANY REQUESTS
5 < {
6 <   "error": "rate limit exceeded"
7 < }
```